Python Programming Assignment

Topics Covered: Flowcharts, Algorithms, Variables, Data Types, User Input, Operators, Arithmetic Logic, Simple Programs, if-else Conditions, Mini Login System

This assignment is designed to help students strengthen their understanding of basic Python programming concepts through practical implementation. Students are expected to apply logical thinking, problem-solving techniques, and programming fundamentals while completing the questions.

Section 1: Flowcharts & Algorithms

Flowcharts and algorithms help programmers understand the logical sequence of steps required to solve a problem.

Question 1: ATM Withdrawal Algorithm & Flowchart

Create a detailed algorithm and flowchart for an ATM withdrawal system. **Requirements:** Input account balance from the user Input withdrawal amount Check whether the withdrawal amount is less than or equal to the available balance If balance is sufficient: Display “Transaction Successful” Display remaining balance Otherwise display “Insufficient Balance” **Expected Output:**
A properly designed algorithm and a flowchart using standard symbols.

Question 2: Student Result Flowchart

Create an algorithm and flowchart for checking student results. **Requirements:** Input student marks If marks are greater than or equal to 35, display “Pass” Otherwise display “Fail” Add grading conditions: 90 and above → Grade A 75–89 → Grade B 50–74 → Grade C Below 50 → Fail **Learning Objective:**
Students will understand decision-making using flowcharts and conditions.

Section 2: Variables & Data Types

Variables are used to store data in a program. Python supports multiple data types such as integers, floats, strings, and booleans.

Question 1: Student Information Program

Write a Python program that stores and displays student details. **Requirements:** Store student name Store age Store percentage Store pass status (True/False) Display all details Use type() function to display datatype of each variable **Sample Variables:**

name = "Rahul"

age = 20

percentage = 85.5

passed = True

Question 2: Variable Naming Rules

Create a list of valid and invalid variable names. **Requirements:** Write 5 valid variable names Write 5 invalid variable names Explain why the invalid variable names are incorrect **Example:**

Valid: student_name, total_marks

Invalid: 1name, student-name

Section 3: User Input

User input allows programs to become interactive and dynamic.

Question 1: Personal Introduction Program

Create a Python program that takes user details as input. **Requirements:** Input name Input age Input city Display a formatted introduction message **Sample Output:**

Hello Rahul

You are 20 years old

You live in Mumbai

Question 2: Simple Interest Calculator

Write a Python program to calculate Simple Interest. **Formula:**

SI = (P × R × T) / 100 **Requirements:** Take Principal amount as input Take Rate of Interest as input Take Time as input Display calculated Simple Interest **Sample Output:**

Simple Interest = 1000

Section 4: Operators & Arithmetic Logic

Operators are symbols used to perform mathematical and logical operations in Python.

Question 1: Arithmetic Calculator

Create a calculator program. **Requirements:** Take two numbers as input Perform: Addition Subtraction Multiplication Division Modulus Display all results properly

Question 2: Driving License Eligibility Checker

Write a Python program using logical operators. **Conditions:** User age must be greater than or equal to 18 User must have valid ID proof If both conditions are true, display “Eligible” Otherwise display “Not Eligible” **Concepts Used:**
and, or, not operators

Section 5: Simple Programs

Simple programs help students apply programming fundamentals in practical situations.

Question 1: Even or Odd Checker

Write a Python program that checks whether a number is even or odd. **Requirements:** Take a number as input Use modulus operator (%) If remainder is 0, display “Even Number” Otherwise display “Odd Number”

Question 2: Area of Rectangle

Write a Python program to calculate the area of a rectangle. **Formula:**
Area = Length × Breadth **Requirements:** Input length Input breadth Calculate area Display the result

Section 6: if-else Conditions

Conditional statements help programs make decisions based on user input and conditions.

Question 1: Positive, Negative, or Zero

Write a Python program to check the nature of a number. **Requirements:** Take a number from the user If number is greater than 0 → Positive If number is less than 0 → Negative If number is equal to 0 → Zero

Question 2: Grade Calculator

Create a grade calculator using if-elif-else conditions. **Requirements:** Input marks from the user Display grades according to conditions: 90+ → Grade A 75–89 → Grade B 50–74 → Grade C Below 50 → Fail

Section 7: Mini Login System

Login systems are one of the most common real-world uses of conditional statements.

Question 1: Basic Login System

Create a simple login system using Python. **Requirements:** Store correct username and password Take username and password as input from user If both are correct: Display “Login Successful” Otherwise: Display “Invalid Username or Password”

Question 2: Advanced Login System

Enhance the login system by adding more features. **Requirements:** Allow maximum 3 login attempts Display failed attempt messages After 3 wrong attempts display “Account Locked” Add password length validation Display a greeting message for admin users **Bonus Challenge:**
Add a secret admin code for special access.

Learning Outcome:

After completing this assignment, students will be able to: Understand algorithms and flowcharts Use variables and data types effectively Take user input in Python Perform arithmetic and logical operations Apply conditional statements Create simple real-world applications Improve problem-solving and logical thinking skills